

Model	Method	Cancer-Myth		CREPE		Medical QA			General QA	
		FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	MQ ↑	PQ ↑	MB ↑	MMLU ↑	GSM ↑
Qwen2.5 7B	CoT	—	—	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	Always correct	—	—	—	—	—	—	—	—	—
	Direct-gated response	—	—	—	—	—	—	—	—	—
	CoT-gated response	—	—	—	—	—	—	—	—	—
	Text-gated response	—	—	—	—	—	—	—	—	—
GPT-6 Luna	CoT	—	—	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	Always correct	—	—	—	—	—	—	—	—	—
	Direct-gated response	—	—	—	—	—	—	—	—	—
	CoT-gated response	—	—	—	—	—	—	—	—	—
	Text-gated response	—	—	—	—	—	—	—	—	—
Gemma 4 12B	CoT	—	—	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	Always correct	—	—	—	—	—	—	—	—	—
	Direct-gated response	—	—	—	—	—	—	—	—	—
	CoT-gated response	—	—	—	—	—	—	—	—	—
	Text-gated response	—	—	—	—	—	—	—	—	—
Qwen3.8 27B OFF	CoT	—	—	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	Always correct	—	—	—	—	—	—	—	—	—
	Direct-gated response	—	—	—	—	—	—	—	—	—
	CoT-gated response	—	—	—	—	—	—	—	—	—
	Text-gated response	—	—	—	—	—	—	—	—	—
Qwen3.8 27B ON	CoT	—	—	—	—	—	—	—	—	—
	Balanced instruction	—	—	—	—	—	—	—	—	—
	Always correct	—	—	—	—	—	—	—	—	—
	Direct-gated response	—	—	—	—	—	—	—	—	—
	CoT-gated response	—	—	—	—	—	—	—	—	—
	Text-gated response	—	—	—	—	—	—	—	—	—

Table S2. Additional response baselines

Unmeasured plan: — is not a measured score. All metrics are percentages.
FPQ: correction; NFP/TPQ: normal-question preservation (Well ≥ 4).
MQ: MedQA; PQ: PubMedQA; MB: Medbullets; GSM: GSM8K. No cross-dataset average.
General QA: MMLU accuracy and GSM8K final-answer exact match; models use the same test protocol.